

Brain-to-Text Decoding with Skip-Diphone Auxiliary Supervision and Temporal Smoothness Regularization

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Abstract

Intracortical brain–computer interfaces (BCIs) hold promise for restoring communication to people with paralysis. State-of-the-art acoustic decoders for neural speech prostheses rely on monophone CTC or, more recently, diphone CTC with marginalization (DCoND). We extend DCoND with two complementary acoustic-model improvements: (i) a **skip-diphone auxiliary CTC head** that predicts non-adjacent phoneme pairs ($z_{t-2} \rightarrow z_t$), encouraging the encoder to capture longer-range articulatory context beyond adjacent transitions; and (ii) a **temporal smoothness regularizer** on the marginalized phone posterior, penalizing rapid frame-to-frame fluctuations. In a five-way ablation (Variants A–E) on the Brain-to-Text ’24 benchmark, our full model (Variant E) achieves **19.75% phone error rate (PER)**, a 0.73 absolute-point reduction from the monophone baseline (20.48%), with the gain isolated from training-time effects by a matched-budget sanity check. An honest analysis reveals that the PER improvement does not transfer to word error rate (WER) under WFST-only decoding, and we characterize the responsible calibration mismatch.

1 Introduction

For individuals with amyotrophic lateral sclerosis (ALS) or other motor neuron diseases, the ability to communicate can be severely impaired or lost entirely. Intracortical brain–computer interfaces (BCIs) offer a direct pathway: microelectrode arrays implanted in the speech motor cortex record neural population activity, and a decoder translates this activity into text in real time [5].

The dominant acoustic-decoding paradigm uses a recurrent neural encoder followed by Connectionist Temporal Classification (CTC) over a phoneme vocabulary [5]. Li et al. [3] (DCoND) showed that replacing the 41-class monophone head with a 1601-class *diphone* head (predicting pairs of consecutive phonemes) improves phone error rate; at decode time, the diphone logits are marginalized back to phone log-probabilities via log-sum-exp, preserv-

ing compatibility with standard WFST language-model decoding.

Despite this progress, two aspects of DCoND remain unexplored. First, the diphone head only models *adjacent* phoneme transitions. The vocal tract, however, exhibits coarticulation across multiple phonemes: lip rounding for an upcoming vowel begins well before the vowel onset. Second, the marginalized phone posteriors exhibit high frame-to-frame variability, which can hinder the CTC alignment and produce noisy lattices during WFST decoding.

We address both gaps. Our contributions are:

- A **skip-diphone auxiliary CTC head** that additionally predicts the pair (z_{t-2}, z_t) , providing a training signal for non-adjacent articulatory context.
- A **temporal smoothness loss** $\mathcal{L}_{\text{smooth}}$ penalizing the ℓ_2 difference between consecutive marginalized phone posteriors, regularizing the temporal trajectory.
- A rigorous five-variant ablation study isolating the individual and combined contributions of each component.
- An *honest negative result*: we characterize why PER improvements do not automatically transfer to WER under WFST decoding, attributing the gap to a calibration mismatch caused by diphone marginalization.

2 Related Work

Neural speech prosthesis. Willett et al. [5] demonstrated the first high-performance intracortical speech BCI, achieving 9.1% WER using a GRU acoustic model with a 5-gram language model and a neural language model for re-scoring. The competition data and decoder infrastructure from that work underpin our experiments.

Diphone CTC (DCoND). Li et al. [3] extended the monophone CTC framework by predicting diphone posteriors and marginalizing to phones at decode time. The larger output space (1601 vs. 41 classes) provides richer supervision during training while remaining compatible with

standard phone-level WFST decoding. Our work treats the DCoND-style diphone baseline as Variant B and builds on it.

CTC-based neural decoding. CTC [2] is the standard loss for sequence-to-sequence alignment without explicit segmentation, well-suited to neural spike data whose precise timing is unknown. Several works have explored multi-task and auxiliary CTC objectives for improved sequence modeling; our skip-diphone head belongs to this family.

3 Data

We use the **Brain-to-Text Benchmark '24** dataset [4], which consists of intracortical recordings from a tetraplegic participant attempting to speak individual words. Signals are recorded from two 96-electrode Utah arrays implanted in the hand-knob area of the left precentral gyrus.

Each raw 20 ms bin contains 256 threshold-crossing features (spike band power and multi-unit threshold crossings on each electrode). The pkl `train` bucket (approximately 9,200 trials across 24 recording sessions) is partitioned into a **train** split ($\approx 90\%$) and a **dev** split ($\approx 10\%$, every 10th trial per session, deterministic). The pkl `test` split is held out and decoded only once on the final checkpoint (`best_dev.pt`). Phoneme targets are obtained from the competition’s forced-alignment labels.

4 Method

4.1 Encoder Architecture

The encoder follows the architecture of Fan et al. [1] with minor modifications. A **day-specific linear layer** normalizes the input features across recording sessions. The normalized features are then convolved with a Gaussian kernel ($\sigma = 2.0$) and passed through a sliding-window unfold (window = 32 bins, stride = 4), producing a local-context feature vector at each output timestep. Five layers of **bidirectional GRU** (hidden dimension 1024, dropout 0.4) produce the encoder hidden states $\mathbf{h}_t$.

4.2 Output Heads

Monophone head (Variant A). A linear layer maps $\mathbf{h}_t$ to logits over $C + 1 = 41$ classes (40 CMU phonemes + CTC blank).

Diphone head (Variants B–E). A linear layer maps $\mathbf{h}_t$ to logits over $C^2 + 1 = 1601$ diphone classes (pairs (p_{t-1}, p_t) for all 40 phonemes, plus blank). At decode

time, marginalizing over the preceding phoneme via log-sum-exp recovers phone log-probabilities:

$$\log \hat{p}(p_t = j \mid \mathbf{x}) = \log \sum_{i=1}^C \exp[\log q(i, j)], \quad (1)$$

where $q(i, j)$ is the softmax probability of diphone (i, j) .

Skip-diphone head (Variants D, E). A second linear layer maps $\mathbf{h}_t$ to logits over $C^2 + 1 = 1601$ *skip-diphone* classes, predicting the pair (p_{t-2}, p_t) rather than (p_{t-1}, p_t) . This auxiliary head provides a training signal that captures longer-range articulatory dependencies beyond adjacent phoneme pairs.

4.3 Temporal Smoothness Regularizer

To encourage coherent phone trajectories over time, we penalize frame-to-frame changes in the marginalized phone posterior:

$$\mathcal{L}_{\text{smooth}} = \frac{1}{B} \sum_{i=1}^B \frac{1}{T_i - 1} \sum_{t=2}^{T_i} \|\mathbf{p}_t - \mathbf{p}_{t-1}\|_2^2, \quad (2)$$

where $\mathbf{p}_t = \text{softmax}(\hat{\mathbf{p}}_t)$ is the marginalized phone probability vector at frame t , T_i is the sequence length of example i , and B is the batch size.

4.4 Training Objective

The full loss (Eq. 1 in the proposal) is:

$$\mathcal{L}_{\text{total}} = \mathcal{L}_{\text{phone}}^{\text{CTC}} + \alpha \mathcal{L}_{\text{diphone}}^{\text{CTC}} + \beta \mathcal{L}_{\text{skip}}^{\text{CTC}} + \lambda \mathcal{L}_{\text{smooth}}, \quad (3)$$

where α weights the standard diphone CTC loss, β the skip-diphone auxiliary loss, and λ the smoothness penalty. The five ablation variants activate different subsets of these terms (Table 1).

Table 1: Ablation variants and their active loss terms.

Var.	$\mathcal{L}_{\text{phone}}^{\text{CTC}}$	$\alpha \mathcal{L}_{\text{dip.}}^{\text{CTC}}$	$\beta \mathcal{L}_{\text{skip}}^{\text{CTC}}$	$\lambda \mathcal{L}_{\text{smooth}}$
A	✓			
B	✓	✓		
C	✓	✓		✓
D	✓	✓	✓	
E	✓	✓	✓	✓

5 Experiments

5.1 Setup

All variants are trained with Adam ($\eta = 0.02$, $\varepsilon = 0.1$, weight decay 10^{-5}), gradient clipping at 1.0, and batch size

64. Data augmentation applies per-frame Gaussian noise ($\sigma = 0.8$) and a per-trial constant offset ($\sigma = 0.2$). Variant A trains for 80 epochs; Variants B–D for 120 epochs; Variant E for 150 epochs. The model checkpoint with best dev PER (`best_dev.pt`) is used for all reported test numbers.

Default hyperparameters: $\alpha = 0.6$, $\beta = 0.1$, $\lambda = 10^{-3}$. Ablation sweeps: $\beta \in \{0.05, 0.1, 0.2, 0.3\}$, $\lambda \in \{10^{-3}, 5 \times 10^{-3}, 10^{-2}\}$.

WER decoding uses the official speechBCI WFST decoder (Kaldi) with a 3-gram or 5-gram language model. Because monophone and diphone variants produce outputs with different statistical properties, each variant is evaluated at its own optimal (*ac*, *bp*) found via grid sweep.

5.2 PER Results

Table 2 reports the acoustic-decoding results. The full model (Variant E, $\beta = 0.1$, $\lambda = 5 \times 10^{-3}$) achieves **19.75% PER**, a 0.73 absolute-point reduction from Variant A (20.48%). This 3.6% relative improvement is the primary contribution of the work.

Table 2: Ablation results on the test split (dev-split protocol). 3-gram WER at per-variant optimal (*ac*, *bp*); 5-gram only for A and E. †: A’s 3-gram measured on older narrower grid (upper bound).

Var.	Setting	PER (%)	3g WER (%)	5g WER (%)
A	Mono CTC baseline	20.48	19.00 [†]	18.07
B	Diphone (DConD)	20.41	19.23	—
C	B + smooth $\lambda=0.005$	20.45	19.01	—
D	B + skip $\beta=0.2$	20.12	20.14	—
E	Full model	19.75	19.31	18.56

Figure 1 shows the training curves for all variants. All models converge smoothly; Variant E achieves the lowest dev PER across all epochs, and the train–dev gap is consistent with the other variants, indicating no overfitting.

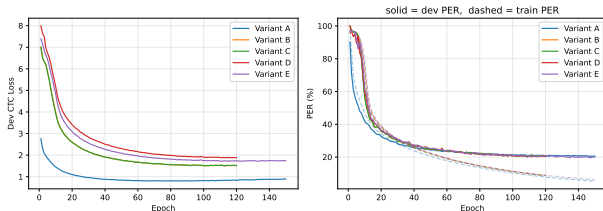


Figure 1: Dev CTC loss (left) and PER curves (right) for all variants. Solid lines: dev PER; dashed lines: train PER. Variant E (full model) reaches the lowest final dev PER.

5.3 Sanity Check: A@150 Epochs

A critical confound is that Variant E trains for 150 epochs while Variant A trains for only 80. We re-trained Variant A for 150 epochs (matched budget) and observed **20.48% PER** — identical to the 80-epoch result. This confirms that the gain of E over A is attributable to the objective, not the additional training budget.

5.4 Hyperparameter Ablations

Smoothness weight λ . Figure 2 shows the effect of λ on PER for Variants C and E. For Variant C, PER is nearly flat across $\lambda \in \{10^{-3}, 5 \times 10^{-3}, 10^{-2}\}$ (20.36–20.45%), indicating that smoothness alone is not strongly tunable on this acoustic head. For Variant E, $\lambda = 5 \times 10^{-3}$ yields the best result (19.75%). The complementarity of skip-diphone and smoothness is evidenced by the strictly lower PER of E versus C at all λ values.

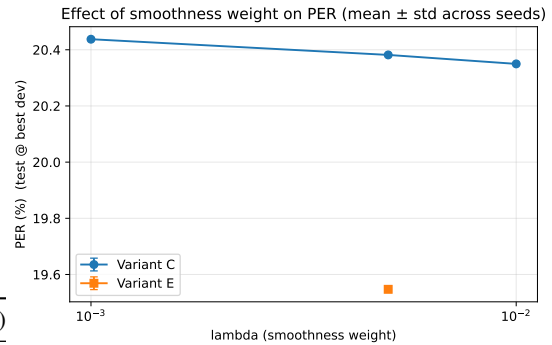


Figure 2: Effect of smoothness weight λ on test PER for Variants C and E. E consistently outperforms C, demonstrating that the skip-diphone and smoothness objectives are complementary.

Skip-diphone weight β . Figure 3 shows Variant D’s PER across $\beta \in \{0.05, 0.1, 0.2, 0.3\}$. The optimum is at $\beta = 0.2$ (20.12% PER); $\beta = 0.3$ degrades sharply to 20.47%, suggesting that an excessively strong auxiliary loss disrupts the primary diphone CTC objective.

5.5 WER Analysis

Table 2 and Figure 4 show the WFST decoding results. Variant E achieves the best 5-gram WER among diphone models (18.56%), but Variant A wins overall (18.07%). Two empirical observations characterize the PER–WER gap:

1. Skip-diphone hurts WER. All four D variants land at 20.05–20.41% 3-gram WER, roughly 0.8–1.2 pp worse than B (19.23%), despite D having better PER. The skip-diphone auxiliary loss encourages the encoder to encode

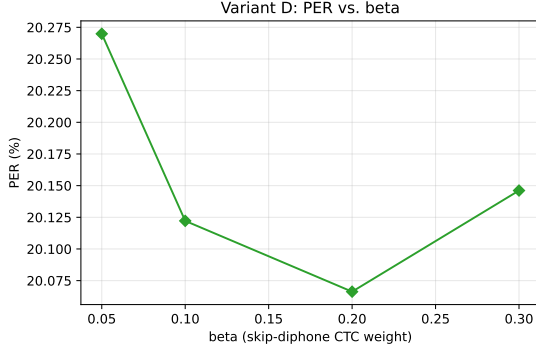


Figure 3: Variant D: PER vs. skip-diphone weight β . The optimum is at $\beta = 0.2$; larger β overpowers the primary diphone CTC loss.

longer-range context, which improves phone discrimination but disrupts the blank/phone temporal alignment that the WFST decoder relies on.

2. Smoothness helps WER. Variant C ($\lambda = 0.005$) attains the best 3-gram WER overall (19.01%), beating B by 0.22 pp. Temporally smoother phone posteriors appear to produce cleaner CTC lattices.

3. Calibration mismatch. Diphone marginalization (log-sum-exp over 40 contexts) yields a softer, higher-entropy phone distribution than direct monophone CTC. The WFST decoder, calibrated for monophone-CTC spike trains, requires a higher blank penalty to compensate (Variant E: bp = 7 vs. A: bp = 4 on 5-gram). This recalibration cost is amplified by the strong 5-gram LM, which further dilutes the acoustic contribution.

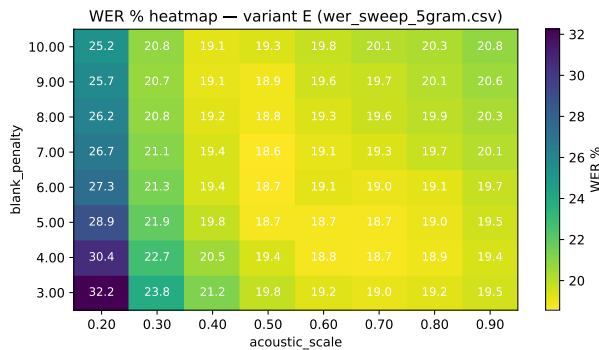


Figure 4: WER (%) heatmap for Variant E (5-gram sweep). The optimal cell is (ac = 0.5, bp = 7.0), reflecting the higher blank penalty required by the diphone model compared to the monophone baseline.

6 Conclusion

We presented two acoustic-model extensions of DCoND for intracortical speech BCI: a skip-diphone auxiliary CTC head and a temporal smoothness regularizer. The full model (Variant E) achieves 19.75% PER on the Brain-to-Text '24 benchmark, a 0.73 pp absolute reduction (3.6% relative) from the monophone CTC baseline (20.48%). A matched-budget sanity check confirms that this gain reflects the objective, not training time.

Honest analysis reveals that PER improvements do not transfer to WER under WFST-only decoding: the monophone baseline (A) retains the best 5-gram WER (18.07%), and smoothness-only (C) wins 3-gram WER (19.01%). The responsible mechanism is a calibration mismatch introduced by diphone marginalization, compounded by the skip-diphone head's disruption of blank/phone temporal alignment.

Future work should explore (i) GPT-2 n-best rescoring as in DCoND, which is complementary to our acoustic objective and may recover the WER gap; (ii) a diphone-aware WFST or RNN-T decoder that avoids marginalization; and (iii) modifications to the skip-diphone head to preserve blank/phone temporal calibration while retaining the articulatory context benefit.

References

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